

Appendix

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A.1 Convex Functions.

Here, we prove the assertions about convex functions used in Theorem 7.46(v).

Definition a.1. A real valued function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a *convex function* if for any $p \in (0, 1)$ and $x, y \in \mathbb{R}$, the following inequality holds.

$$\varphi(px + (1 - p)y) \leq p\varphi(x) + (1 - p)\varphi(y).$$

For such a convex function and for any $x, y \in \mathbb{R}$, $x \neq y$, let the *secant slope* of the line between $(x, \varphi(x))$ and $(y, \varphi(y))$ be denoted by

$$S_{x,y} \triangleq \frac{\varphi(y) - \varphi(x)}{y - x}.$$

Clearly $S_{x,y} = S_{y,x}$ for any $x, y \in \mathbb{R}$, $x \neq y$.

Theorem a.2. Let φ be a convex function and let $x < y < z$ be three real numbers. Then $S_{x,y} \leq S_{x,z} \leq S_{y,z}$.

Cru. Let $y = px + (1 - p)z$ with $p = \frac{z-y}{z-x} \in (0, 1)$. The defining convexity property $\varphi(y) \leq p\varphi(x) + (1 - p)\varphi(z)$ algebraically rearranges directly to:

$$\frac{\varphi(y) - \varphi(x)}{y - x} \leq \frac{\varphi(z) - \varphi(x)}{z - x} \leq \frac{\varphi(z) - \varphi(y)}{z - y}$$

which is identically $S_{x,y} \leq S_{x,z} \leq S_{y,z}$. □

Definition a.3. Let φ be a convex function and let $x_0 \in \mathbb{R}$. We denote the left hand derivative of φ at x_0 by $\varphi'_-(x_0)$ and the right hand derivative of φ at x_0 by $\varphi'_+(x_0)$ whenever they exist.

Theorem a.4. Let φ be a convex function. Then for every $x_0 \in \mathbb{R}$, $\varphi'_-(x_0)$ and $\varphi'_+(x_0)$ exist and $\varphi'_-(x_0) \leq \varphi'_+(x_0)$. Also, for any $x, y \in \mathbb{R}$ such that $x < y$, $\varphi'_+(x) \leq S_{x,y} \leq \varphi'_-(y)$.

CruX. From Theorem a.2, for any $x^- < x_0 < x^+$, $S_{x^-,x_0} \leq S_{x_0,x^+}$. Taking the limit $x^- \uparrow x_0$ yields $\varphi'_-(x_0) \leq S_{x_0,x^+}$. Taking the limit $x^+ \downarrow x_0$ then yields $\varphi'_-(x_0) \leq \varphi'_+(x_0)$.

Next, let $x < z < y$. Theorem a.2 guarantees $S_{x,z} \leq S_{x,y} \leq S_{z,y}$. Taking the limit $z \downarrow x$ on the left inequality yields $\varphi'_+(x) \leq S_{x,y}$. Taking the limit $z \uparrow y$ on the right inequality yields $S_{x,y} \leq \varphi'_-(y)$. $\square$

Corollary a.5. Any convex function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable m -a.e. (here, m represents the Lebesgue measure on the set of real numbers).

CruX. If φ is non-differentiable at $x_0 \in \mathbb{R}$, then $\varphi'_-(x_0) < \varphi'_+(x_0)$. We can choose a rational number $q_{x_0} \in (\varphi'_-(x_0), \varphi'_+(x_0))$. If $x_0 < y_0$ are two points of non-differentiability, Theorem a.4 ensures $q_{x_0} < \varphi'_+(x_0) \leq \varphi'_-(y_0) < q_{y_0}$ where q_{y_0} is the corresponding rational number chosen from $(\varphi'_-(y_0), \varphi'_+(y_0))$. Because the assigned rational numbers q_{x_0} and q_{y_0} are strictly ordered and distinct for any distinct points of non-differentiability, the set of such points must be countable. A countable set is an m -null set, so φ is differentiable m -a.e. $\square$

Definition a.6. Locally Lipschitz Convex Function. A convex function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is said to be *locally Lipschitz* if for every $x \in \mathbb{R}$, there exists an open interval $(a, b) \subset \mathbb{R}$ containing x and a constant $L_x \geq 0$ such that

$$|\varphi(y) - \varphi(z)| \leq L_x |y - z| \quad \text{for all } y, z \in (a, b).$$

Since for any locally Lipschitz function φ , and for any $x \in \mathbb{R}$, we have $\lim_{y \rightarrow x} |\varphi(y) - \varphi(x)| \leq \lim_{y \rightarrow x} L_x |y - x| = 0$, such functions are clearly continuous everywhere.

Corollary a.7. Any convex function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is locally Lipschitz.

CruX. Let $x \in \mathbb{R}$ and choose an interval (a, b) containing x . For any $y, z \in (a, b)$ with $y < z$, Theorem a.4 implies $\varphi'_+(a) \leq S_{a,y} \leq S_{y,z} \leq S_{z,b} \leq \varphi'_-(b)$. Therefore, the secant slope $S_{y,z}$ is strictly bounded in absolute value by $L_x = \max(|\varphi'_+(a)|, |\varphi'_-(b)|)$. This implies $|\varphi(y) - \varphi(z)| \leq L_x |y - z|$, proving local Lipschitz continuity. $\square$

A.2 Dynkin's $\pi - \lambda$ Theorem.

In Chapter 6, the monotone class theorem was developed which helps in establishing that measures that agree on sets from a specific generating class of sets, namely a *field* (note that a field is also referred to as an *algebra* in many texts) will agree on the σ -field that it generates. Another tool that comes in handy is Dynkin's $\pi - \lambda$ Theorem.

Definition a.8. A class $\mathcal{C}$ of subsets of Ω is a π -system (or a π -class) if

$$A, B \in \mathcal{C} \implies A \cap B \in \mathcal{C}.$$

Definition a.9. A class $\mathcal{L}$ of subsets of Ω is a λ -system (or a λ -class) if:

1. $\Omega \in \mathcal{L}$,
2. $A, B \in \mathcal{L}, A \subset B \implies B \setminus A \in \mathcal{L}$, and
3. $A_n \in \mathcal{L}, A_n \subset A_{n+1}$ for all $n \geq 1 \implies \bigcup_{n \geq 1} A_n \in \mathcal{L}$.

Remark on the Relationship between λ -systems and Monotone Classes: Every λ -system is a monotone class, but a monotone class is not necessarily a λ -system mainly due to the presence of Ω in λ -systems while monotone classes need not contain Ω .

For any $\mathcal{B} \subset \mathcal{P}(\Omega)$, the **λ -system generated by $\mathcal{B}$** , denoted by $\lambda(\mathcal{B})$, is defined as the intersection of all λ -systems containing $\mathcal{B}$. It is the smallest λ -system containing $\mathcal{B}$, because the intersection of any family of λ -systems containing $\mathcal{B}$ is also a λ -system containing $\mathcal{B}$.

Theorem a.10 (The π - λ Theorem). *If $\mathcal{C}$ is a π -system, then $\lambda(\mathcal{C}) = \sigma(\mathcal{C})$.*

Cruux. Clearly every σ -field is a λ -system so $\lambda(\mathcal{C}) \subseteq \sigma(\mathcal{C})$. Since $\lambda(\mathcal{C})$ is a λ -system, it is closed under complementation and monotone increasing unions. If we had closure under finite intersections, then closure under finite unions is immediate, turning $\lambda(\mathcal{C})$ into a field closed under increasing limits. In the proof of Lemma 6.7, it was shown that a field closed under increasing limits is a σ -field, so all we need to do is show that $\lambda(\mathcal{C})$ is a π -system. Let

$$\lambda_1(\mathcal{C}) \triangleq \{A \in \lambda(\mathcal{C}) : A \cap B \in \lambda(\mathcal{C}) \ \forall B \in \mathcal{C}\}.$$

We can verify $\lambda_1(\mathcal{C})$ is a λ -system because set intersection distributes over proper differences (e.g., $(A_2 \setminus A_1) \cap B = (A_2 \cap B) \setminus (A_1 \cap B)$) and over countable increasing unions. Since $\mathcal{C}$ is a π -system, $\mathcal{C} \subset \lambda_1(\mathcal{C})$. Minimality implies $\lambda(\mathcal{C}) \subset \lambda_1(\mathcal{C})$; hence $\lambda_1(\mathcal{C}) = \lambda(\mathcal{C})$. Next, let

$$\lambda_2(\mathcal{C}) \triangleq \{A \in \lambda(\mathcal{C}) : A \cap B \in \lambda(\mathcal{C}) \ \forall B \in \lambda(\mathcal{C})\}.$$

By the exact same distributive properties, $\lambda_2(\mathcal{C})$ is also a λ -system. From the previous step, $\mathcal{C} \subset \lambda_2(\mathcal{C})$. Since $\lambda_2(\mathcal{C})$ is a λ -system, $\lambda(\mathcal{C}) \subset \lambda_2(\mathcal{C})$, forcing $\lambda_2(\mathcal{C}) = \lambda(\mathcal{C})$. Thus $\lambda(\mathcal{C})$ is closed under finite intersections, making it a σ -algebra ($\sigma(\mathcal{C})$). $\square$

Similar to the uniqueness argument in Theorem 6.8, we can use π - λ Theorem to prove uniqueness of measures.

Definition a.11 (Pre-measure). Let $\mathcal{P}$ be a family of subsets of a sample space Ω containing the empty set $\emptyset$. A *pre-measure* on $\mathcal{P}$ is a set function $\mu : \mathcal{P} \rightarrow [0, \infty]$ satisfying the following properties:

- (i) $\mu(\emptyset) = 0$.
- (ii) If $\{A_n\}_{n \geq 1}$ is a sequence of mutually disjoint sets in $\mathcal{P}$ such that their union $\bigcup_{n \geq 1} A_n$ also belongs to $\mathcal{P}$, then

$$\mu \left(\bigcup_{n \geq 1} A_n \right) = \sum_{n \geq 1} \mu(A_n).$$

Definition a.12 (Semialgebra). Let Ω be a nonempty set. A family of sets $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ is called a *semialgebra* if it satisfies the following properties:

- (i) $\emptyset \in \mathcal{C}$.
- (ii) If $A, B \in \mathcal{C}$, then $A \cap B \in \mathcal{C}$ (closed under finite intersections).
- (iii) For any $A \in \mathcal{C}$, there exist mutually disjoint sets $B_1, B_2, \dots, B_k \in \mathcal{C}$ for some finite $k \geq 1$ such that

$$A^c = \bigcup_{i=1}^k B_i.$$

Note: A *field* is a semialgebra where $k = 1$ always suffices in condition (iii), meaning the complement of any set is itself an element of the family, making it closed under finite unions. Also note that the family $\mathcal{P}$ in Definition a.11 can be a semialgebra or a field; if it is a σ -field, then the pre-measure is a measure.

Theorem a.13 (Uniqueness Step of Carathéodory's Extension). *Let $\mathcal{A}$ be a field of subsets of Ω , and let μ_0 be a σ -finite pre-measure on $\mathcal{A}$. If μ and ν are two measures on the generated σ -algebra $\sigma(\mathcal{A})$ such that $\mu(A) = \nu(A) = \mu_0(A)$ for all $A \in \mathcal{A}$, then $\mu(E) = \nu(E)$ for all $E \in \sigma(\mathcal{A})$.*

Cruæ. By definition of σ -finiteness, there exists a sequence of disjoint sets $\Omega_n \in \mathcal{A}$ such that $\Omega = \bigcup_{n=1}^{\infty} \Omega_n$ and $\mu_0(\Omega_n) < \infty$ for all $n \geq 1$. Fix $n \geq 1$, and define two finite localized measures μ_n, ν_n on $\sigma(\mathcal{A})$ by

$$\mu_n(E) \triangleq \mu(E \cap \Omega_n) \quad \text{and} \quad \nu_n(E) \triangleq \nu(E \cap \Omega_n).$$

Now define the family of agreement sets for this specific component:

$$\mathcal{L}_n \triangleq \{E \in \sigma(\mathcal{A}) : \mu_n(E) = \nu_n(E)\}.$$

We show $\mathcal{L}_n$ is a λ -system. First, $\Omega \in \mathcal{L}_n$ because $\mu_n(\Omega) = \mu(\Omega_n) = \mu_0(\Omega_n) = \nu(\Omega_n) = \nu_n(\Omega)$. If $A, B \in \mathcal{L}_n$ with $A \subset B$, the finiteness of $\mu_0(\Omega_n)$ justifies subtraction: $\mu_n(B \setminus A) = \mu_n(B) - \mu_n(A) = \nu_n(B) - \nu_n(A) = \nu_n(B \setminus A)$, so $B \setminus A \in \mathcal{L}_n$. Lastly, if $A_k \in \mathcal{L}_n$ and $A_k \uparrow A$, continuity from below of measures yields $\mu_n(A) = \lim \mu_n(A_k) = \lim \nu_n(A_k) = \nu_n(A)$, hence $A \in \mathcal{L}_n$. Thus, $\mathcal{L}_n$ is a λ -system.

Since $\mathcal{A}$ is a field, it is a π -system. For any $A \in \mathcal{A}$, we have $A \cap \Omega_n \in \mathcal{A}$ by field closure. Thus $\mu_n(A) = \mu_0(A \cap \Omega_n) = \nu_n(A)$, proving $\mathcal{A} \subset \mathcal{L}_n$. By the π - λ Theorem, $\sigma(\mathcal{A}) = \lambda(\mathcal{A}) \subset \mathcal{L}_n$, implying $\mu(E \cap \Omega_n) = \nu(E \cap \Omega_n)$ for all $E \in \sigma(\mathcal{A})$. Summing over all partitions and leveraging countably additive disjoint limits, we recover the global identity:

$$\mu(E) = \sum_{n=1}^{\infty} \mu(E \cap \Omega_n) = \sum_{n=1}^{\infty} \nu(E \cap \Omega_n) = \nu(E) \quad \forall E \in \sigma(\mathcal{A}).$$

□

A.3 Carathéodory's Extension Theorem

In Section 2.2, the outer measure was defined in Definition 2.3. In Section 2.3, the definition of *measurability* was given in Definition 2.9, and properties of measurable sets were proven in Theorem 2.11. This entire theory was developed based on coverings by basic intervals. In a general setting, we can abstract this development to other families of sets—specifically, to semialgebras on which a pre-measure is defined.

Lemma a.14. *The smallest field containing a semialgebra $\mathcal{C}$ is given by the family of all finite disjoint unions of sets from $\mathcal{C}$:*

$$\mathcal{A}_{\mathcal{C}} = \left\{ \bigcup_{i=1}^k B_i : B_1, \dots, B_k \in \mathcal{C} \text{ are mutually disjoint, } k \geq 1 \right\}.$$

CruX. We establish that $\mathcal{A}_{\mathcal{C}}$ satisfies the definition of a field:

- **Contains Ω :** Since $\mathcal{C}$ is a semialgebra, $\emptyset \in \mathcal{C}$, meaning $\Omega = \emptyset^c$ is a finite disjoint union of elements in $\mathcal{C}$. Thus, $\Omega \in \mathcal{A}_{\mathcal{C}}$.
- **Intersections:** Let $A = \bigcup_{i=1}^k B_i$ and $D = \bigcup_{j=1}^m C_j$ belong to $\mathcal{A}_{\mathcal{C}}$. Distributing the intersection yields $A \cap D = \bigcup_{i,j} (B_i \cap C_j)$. Since $\mathcal{C}$ is a semialgebra, each $B_i \cap C_j \in \mathcal{C}$. Furthermore, these blocks are mutually disjoint because the original component streams were disjoint. Thus, $A \cap D \in \mathcal{A}_{\mathcal{C}}$.
- **Complements:** Let $A = \bigcup_{i=1}^k B_i \in \mathcal{A}_{\mathcal{C}}$. By De Morgan's Laws, $A^c = \bigcap_{i=1}^k B_i^c$. Each B_i^c is a finite disjoint union of elements in $\mathcal{C}$, meaning $B_i^c \in \mathcal{A}_{\mathcal{C}}$. Because we just proved that $\mathcal{A}_{\mathcal{C}}$ is closed under finite intersections, the finite intersection of these B_i^c terms must also belong to $\mathcal{A}_{\mathcal{C}}$. Hence, $A^c \in \mathcal{A}_{\mathcal{C}}$.
- **Finite Unions:** For any $A, D \in \mathcal{A}_{\mathcal{C}}$, we use the identity $A \cup D = (A^c \cap D^c)^c$. Since $\mathcal{A}_{\mathcal{C}}$ is closed under complements and intersections, it is closed under finite unions.

Thus, $\mathcal{A}_{\mathcal{C}}$ is a field. Finally, since any field containing $\mathcal{C}$ must be closed under finite unions, it must automatically contain all finite disjoint unions of elements from $\mathcal{C}$. Therefore, $\mathcal{A}_{\mathcal{C}}$ is nested inside every field containing $\mathcal{C}$, proving it is the unique smallest field. $\square$

Proposition a.15. *Let μ be a pre-measure on a semialgebra $\mathcal{C}$, and let $\mathcal{A}$ be the smallest field generated by $\mathcal{C}$. For each $A \in \mathcal{A}$, define*

$$\bar{\mu}(A) \triangleq \sum_{i=1}^k \mu(B_i),$$

where $A = \bigcup_{i=1}^k B_i$ is a representation of A as a finite union of mutually disjoint sets $B_1, \dots, B_k \in \mathcal{C}$ for some $k \geq 1$. Then:

(i) $\bar{\mu}$ is well-defined (independent of the disjoint representation of A).

(ii) $\bar{\mu}$ is finitely additive on $\mathcal{A}$.

(iii) $\bar{\mu}$ is countably additive on $\mathcal{A}$.

CruX. (i) Let $A = \bigcup_{i=1}^k B_i = \bigcup_{j=1}^l D_j$ be two different finite disjoint representations with $B_i, D_j \in \mathcal{C}$. For each fixed i , we can write $B_i = B_i \cap A = \bigcup_{j=1}^l (B_i \cap D_j)$. Since $\mathcal{C}$ is a semialgebra, each $B_i \cap D_j \in \mathcal{C}$, and these are mutually disjoint over j . By the finite additivity of the pre-measure μ on $\mathcal{C}$, we have $\mu(B_i) = \sum_{j=1}^l \mu(B_i \cap D_j)$. Summing over i yields:

$$\sum_{i=1}^k \mu(B_i) = \sum_{i=1}^k \sum_{j=1}^l \mu(B_i \cap D_j) = \sum_{j=1}^l \sum_{i=1}^k \mu(B_i \cap D_j).$$

By symmetric logic, $\mu(D_j) = \sum_{i=1}^k \mu(B_i \cap D_j)$. Substituting this into the right side gives $\sum_{i=1}^k \mu(B_i) = \sum_{j=1}^l \mu(D_j)$, proving $\bar{\mu}$ is well-defined.

(ii) Let $A, B \in \mathcal{A}$ be disjoint with representations $A = \bigcup_{i=1}^k B_i$ and $B = \bigcup_{j=1}^l D_j$. Since $A \cap B = \emptyset$, every B_i is disjoint from every D_j . Thus, $A \cup B = (\bigcup_{i=1}^k B_i) \cup (\bigcup_{j=1}^l D_j)$ is a valid finite disjoint representation in $\mathcal{C}$ for the combined set. By definition:

$$\bar{\mu}(A \cup B) = \sum_{i=1}^k \mu(B_i) + \sum_{j=1}^l \mu(D_j) = \bar{\mu}(A) + \bar{\mu}(B).$$

(iii) Let $\{A_n\}_{n \geq 1} \subseteq \mathcal{A}$ be mutually disjoint such that $A \triangleq \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$. Let $A = \bigcup_{i=1}^k B_i$ and $A_n = \bigcup_{j=1}^{k_n} B_{nj}$ be their respective finite disjoint representations in $\mathcal{C}$. For each fixed i , we can decompose $B_i \in \mathcal{C}$ as a countable disjoint union of elements from $\mathcal{C}$:

$$B_i = B_i \cap A = B_i \cap \left(\bigcup_{n=1}^{\infty} \bigcup_{j=1}^{k_n} B_{nj} \right) = \bigcup_{n=1}^{\infty} \bigcup_{j=1}^{k_n} (B_i \cap B_{nj}).$$

Since μ is countably additive on $\mathcal{C}$, we have $\mu(B_i) = \sum_{n=1}^{\infty} \sum_{j=1}^{k_n} \mu(B_i \cap B_{nj})$. Summing over all i and using the absolute convergence of non-negative series to rearrange terms, we get:

$$\bar{\mu}(A) = \sum_{i=1}^k \mu(B_i) = \sum_{n=1}^{\infty} \sum_{j=1}^{k_n} \sum_{i=1}^k \mu(B_i \cap B_{nj}).$$

Conversely, each component B_{nj} can be broken down as $B_{nj} = B_{nj} \cap A = \bigcup_{i=1}^k (B_{nj} \cap B_i)$. By the finite additivity of μ on $\mathcal{C}$, $\mu(B_{nj}) = \sum_{i=1}^k \mu(B_{nj} \cap B_i)$. Collapsing the

inner sums yields the final result:

$$\bar{\mu}(A) = \sum_{n=1}^{\infty} \sum_{j=1}^{k_n} \mu(B_{nj}) = \sum_{n=1}^{\infty} \bar{\mu}(A_n).$$

□

Definition a.16 (Outer Measure). A set function $\mu^* : \mathcal{P}(\Omega) \rightarrow [0, \infty]$ is called an *outer measure* on Ω if it satisfies the following properties:

- (i) $\mu^*(\emptyset) = 0$.
- (ii) $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$ (monotonicity).
- (iii) If $\{A_n\}_{n \geq 1} \subseteq \mathcal{P}(\Omega)$, then

$$\mu^* \left(\bigcup_{n \geq 1} A_n \right) \leq \sum_{n=1}^{\infty} \mu^*(A_n) \quad (\text{countable subadditivity}).$$

Definition a.17 (Induced Outer Measure). Given a pre-measure μ on a semialgebra $\mathcal{C}$, the outer measure *induced by* μ is the set function μ^* defined on $\mathcal{P}(\Omega)$ such that for any $A \subseteq \Omega$, $\mu^*(A) \triangleq \inf Z_A$, where

$$Z_A = \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subseteq \mathcal{C}, A \subseteq \bigcup_{n \geq 1} A_n \right\}.$$

Note: This is an abstraction of Capinski and Kopp's Definition 2.3 to general pre-measures and semialgebras.

Clearly, Definition a.17 is implying that the induced outer measure is indeed an outer measure satisfying the three conditions in Definition a.16. We prove that next.

Proposition a.18. *The induced outer measure as defined in Definition a.17 satisfies the conditions for an outer measure as required by Definition a.16.*

- CruX.** (i) Take the trivial cover $A_n = \emptyset \in \mathcal{C}$ for all $n \geq 1$. Then $\emptyset \subseteq \bigcup_{n \geq 1} A_n$, so $0 \in Z_{\emptyset}$. Since $\mu^* \geq 0$, $\mu^*(\emptyset) = \inf Z_{\emptyset} = 0$.
- (ii) If $A \subseteq B$, every countable cover of B by sets in $\mathcal{C}$ is also a cover of A . Thus, $Z_B \subseteq Z_A$, which yields $\inf Z_A \leq \inf Z_B \implies \mu^*(A) \leq \mu^*(B)$.

- (iii) Let $\{A_n\}_{n \geq 1} \subseteq \mathcal{P}(\Omega)$. If $\sum_{n \geq 1} \mu^*(A_n) = \infty$, subadditivity holds trivially. Assume $\sum_{n \geq 1} \mu^*(A_n) < \infty$. For any $\epsilon > 0$ and each $n \geq 1$, choose a cover $\{B_{nj}\}_{j \geq 1} \subseteq \mathcal{C}$ of A_n such that $\sum_{j \geq 1} \mu(B_{nj}) < \mu^*(A_n) + \frac{\epsilon}{2^n}$.

Since $\bigcup_{n \geq 1} \bigcup_{j \geq 1} B_{nj}$ forms a countable cover of $\bigcup_{n \geq 1} A_n$ by sets in $\mathcal{C}$, and non-negative double series can be rearranged arbitrarily:

$$\mu^*\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n \geq 1} \sum_{j \geq 1} \mu(B_{nj}) < \sum_{n \geq 1} \left(\mu^*(A_n) + \frac{\epsilon}{2^n}\right) = \sum_{n \geq 1} \mu^*(A_n) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, sending $\epsilon \rightarrow 0$ yields the result. $\square$

Definition a.19 (μ^* -Measurability). A set $A \subseteq \Omega$ is said to be μ^* -measurable (or Carathéodory measurable) if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \forall E \subseteq \Omega.$$

Note: This abstracts Definition 2.9 to general outer measures on abstract sample spaces.

The proof of the next theorem is exactly the same as Theorem 2.11 but will be proven for this abstract setting.

Theorem a.20 (Carathéodory's Measurability Theorem). *Let μ^* be an outer measure on Ω as defined in Definition a.16. Let*

$$\mathcal{M} \triangleq \{A \subseteq \Omega : A \text{ is } \mu^*\text{-measurable as defined in Definition a.19.}\}$$

Then:

- (i) $\mathcal{M}$ is a σ -field.
- (ii) μ^* restricted to $\mathcal{M}$ is a measure.
- (iii) $\mathcal{M}$ is complete; that is, if $\mu^*(N) = 0$ and $G \subseteq N$, then $G \in \mathcal{M}$.

Cruz. (i) **Complements and Ω :** Choosing $E = \Omega$ or $E = \emptyset$ in Definition a.19 shows $\Omega \in \mathcal{M}$. The symmetry of $E = (E \cap A) \cup (E \cap A^c)$ ensures that if $A \in \mathcal{M}$, then $A^c \in \mathcal{M}$.

Finite Disjoint Unions: Let $\{A_k\}_{k=1}^n$ be pairwise disjoint sets in $\mathcal{M}$. For any testing set $E \subseteq \Omega$, testing with the measurable set A_1 gives $\mu^*(E) = \mu^*(E \cap A_1) + \mu^*(E \cap A_1^c)$. Testing the remaining piece $E \cap A_1^c$ with the measurable set A_2 gives $\mu^*(E \cap A_1^c) = \mu^*(E \cap A_1^c \cap A_2) + \mu^*(E \cap A_1^c \cap A_2^c) = \mu^*(E \cap A_2) + \mu^*(E \cap A_1^c \cap A_2^c)$ since the sets are disjoint. Proceeding inductively for n sets yields:

$$\mu^*(E) = \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*\left(E \cap \left(\bigcup_{k=1}^n A_k\right)^c\right).$$

Countable Disjoint Unions: Taking the limit as $n \rightarrow \infty$ on both sides, and using the monotonicity of outer measures since $(\bigcup_{k=1}^n A_k)^c \supseteq (\bigcup_{k=1}^\infty A_k)^c$:

$$\mu^*(E) \geq \sum_{k=1}^{\infty} \mu^*(E \cap A_k) + \mu^*\left(E \cap \left(\bigcup_{k=1}^{\infty} A_k\right)^c\right).$$

Applying countable subadditivity to the infinite sum inside the first piece transforms it into:

$$\mu^*(E) \geq \mu^*\left(E \cap \left(\bigcup_{k=1}^{\infty} A_k\right)\right) + \mu^*\left(E \cap \left(\bigcup_{k=1}^{\infty} A_k\right)^c\right).$$

The reverse inequality holds automatically by subadditivity, proving $\bigcup_{k=1}^\infty A_k \in \mathcal{M}$.

General Countable Unions: If $\{A_k\}_{k \geq 1} \subseteq \mathcal{M}$ are not disjoint, we construct the disjoint sequence $F_1 = A_1$, and $F_n = A_n \setminus \left(\bigcup_{k=1}^{n-1} A_k\right)$ for $n \geq 2$. Since $\mathcal{M}$ is closed under finite unions and complements, each $F_n \in \mathcal{M}$. Because $\bigcup_{k=1}^\infty A_k = \bigcup_{k=1}^\infty F_k$, the countable union belongs to $\mathcal{M}$. Thus, $\mathcal{M}$ is a σ -field.

- (ii) **Countable Additivity:** Set the arbitrary testing set E in our disjoint limit inequality to be the union itself, $E = \bigcup_{k=1}^\infty A_k$. The remainder term drops out since $\mu^*(\emptyset) = 0$, leaving $\mu^*\left(\bigcup_{k=1}^\infty A_k\right) = \sum_{k=1}^\infty \mu^*(A_k)$, proving $\mu^*|_{\mathcal{M}}$ is a measure.
- (iii) **Completeness:** Let $\mu^*(N) = 0$ and $G \subseteq N$. By monotonicity, $0 \leq \mu^*(E \cap G) \leq \mu^*(N) = 0 \implies \mu^*(E \cap G) = 0$. Using monotonicity again, $E \cap G^c \subseteq E \implies \mu^*(E \cap G^c) \leq \mu^*(E)$. Adding these together yields:

$$\mu^*(E) \geq 0 + \mu^*(E \cap G^c) = \mu^*(E \cap G) + \mu^*(E \cap G^c).$$

Subadditivity ensures the reverse inequality, so $G \in \mathcal{M}$. □

Theorem a.21 (Carathéodory's Extension Theorem). *Let μ be a pre-measure on a semi-algebra $\mathcal{C}$, let μ^* be the outer measure induced by μ (Definition a.17), and let $\mathcal{M}$ be the σ -field of μ^* -measurable sets (Definition a.19). Then:*

- (i) μ^* is an outer measure.
- (ii) $\mathcal{C} \subseteq \mathcal{M}$.
- (iii) $\mu^*(A) = \mu(A)$ for all $A \in \mathcal{C}$.

Cruz. (i) Immediate from Proposition a.18.

- (ii) Let $A \in \mathcal{C}$. To show $A \in \mathcal{M}$, it suffices by subadditivity to check $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ for an arbitrary testing set $E \subseteq \Omega$. Let $\{A_n\}_{n \geq 1} \subseteq \mathcal{C}$ be any countable cover of E .

Since $\mathcal{C}$ is a semialgebra, $A^c = \bigcup_{i=1}^k B_i$ for a finite collection of mutually disjoint sets $B_i \in \mathcal{C}$. Because $A_n \cap A \in \mathcal{C}$ and $A_n \cap B_i \in \mathcal{C}$, we can partition A_n into mutually disjoint components within $\mathcal{C}$:

$$A_n = (A_n \cap A) \cup (A_n \cap A^c) = (A_n \cap A) \cup \left(\bigcup_{i=1}^k (A_n \cap B_i) \right).$$

Since μ is a pre-measure, it is finitely additive on these disjoint components:

$$\mu(A_n) = \mu(A_n \cap A) + \sum_{i=1}^k \mu(A_n \cap B_i).$$

Summing over all $n \geq 1$ yields:

$$\sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \mu(A_n \cap A) + \sum_{n=1}^{\infty} \sum_{i=1}^k \mu(A_n \cap B_i).$$

Notice that $\{A_n \cap A\}_{n \geq 1} \subseteq \mathcal{C}$ is a countable cover of $E \cap A$, and $\{A_n \cap B_i\}_{n \geq 1, 1 \leq i \leq k} \subseteq \mathcal{C}$ is a countable cover of $E \cap A^c$. By the definition of μ^* as an infimum:

$$\sum_{n=1}^{\infty} \mu(A_n) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Since this lower bound holds firmly for the sum of *every* valid countable cover of E , taking the infimum over all such covers on the left hand side gives:

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c) \implies A \in \mathcal{M}.$$

- (iii) Let $A \in \mathcal{C}$. Using the trivial cover where $A_1 = A$ and $A_n = \emptyset$ for $n \geq 2$, we get $\mu^*(A) \leq \mu(A) + 0 = \mu(A)$.

For the reverse inequality, let $\{A_n\}_{n \geq 1} \subseteq \mathcal{C}$ be any countable cover of A . By Proposition a.15, we can uniquely extend the pre-measure μ on $\mathcal{C}$ to a countably additive pre-measure $\bar{\mu}$ on the generated field $\mathcal{A}_{\mathcal{C}}$. Within the field $\mathcal{A}_{\mathcal{C}}$, we can safely apply the classic disjointification trick: define $F_1 = A_1$ and $F_n = A_n \setminus \left(\bigcup_{i=1}^{n-1} A_i \right) \in \mathcal{A}_{\mathcal{C}}$.

Since $\bar{\mu}$ is a pre-measure on $\mathcal{A}_{\mathcal{C}}$ and $A = \bigcup_{n=1}^{\infty} (A \cap F_n)$ is a disjoint union:

$$\mu(A) = \bar{\mu}(A) = \sum_{n=1}^{\infty} \bar{\mu}(A \cap F_n) \leq \sum_{n=1}^{\infty} \bar{\mu}(F_n) \leq \sum_{n=1}^{\infty} \bar{\mu}(A_n) = \sum_{n=1}^{\infty} \mu(A_n).$$

Taking the infimum over all covers $\{A_n\}_{n \geq 1}$ on the right-hand side yields $\mu(A) \leq \mu^*(A)$. Combining both bounds gives $\mu^*(A) = \mu(A)$. $\square$

Concluding Remark (The Full Extension Theorem): Theorem *a.21* establishes the *existence* of a measure extension from a semialgebra $\mathcal{C}$ to $\sigma(\mathcal{C})$ via the induced outer measure. If the original pre-measure μ on $\mathcal{C}$ is additionally assumed to be σ -finite, its unique extension $\bar{\mu}$ to the field $\mathcal{A}_{\mathcal{C}}$ (via Proposition *a.15*) remains σ -finite. Therefore, by Theorem *a.13*, the final extended measure on $\sigma(\mathcal{C})$ is entirely *unique*.

Note that the construction of the σ -field $\mathcal{M}$ in Theorem *a.20* kind of overdoes the job of constructing the σ -field. In other words, it is not necessary that $\sigma(\mathcal{C}) = \mathcal{M}$ even though it is a fact that $\sigma(\mathcal{C}) \subseteq \mathcal{M}$. Hence the measure obtained through the extension in *a.21* works well on the measurable space $(\Omega, \sigma(\mathcal{C}))$ but $\sigma(\mathcal{C})$ may not be complete, whereas $\mathcal{M}$ is complete.

Examples: Following are pairs of semialgebra $\mathcal{C}$ and pre-measure μ , where we can easily invoke the results above matching the corresponding results in the text.

- $\mathcal{C} = \{I : I \subseteq \mathbb{R}, I \text{ is an interval.}\}$ The pre-measure $\mu = l$, the length of the interval. Resulting σ -field is the Lebesgue σ -field $\mathcal{M}$. Note that $\sigma(\mathcal{C}) = \mathcal{B}$, the Borel σ -field on which the measure is valid, but is not complete.
- $\mathcal{C} = \{I : I \subseteq \mathbb{R}, I = (a, b] \text{ where } a \in [-\infty, \infty), \text{ or } I = (b, \infty), |b| < \infty \text{ in both cases.}\}$ The pre-measure $\mu = l_F$, the 'generalized interval length' as defined in Section 7.3. Resulting σ -field is the Lebesgue-Stieltjes σ -field $\mathcal{M}_F$. Note that $\sigma(\mathcal{C}) = \mathcal{B}$, the Borel σ -field even though we start with a different semialgebra. As before, $\mathcal{B}$ is not complete but the measure from extension works on the measurable space $(\Omega, \sigma(\mathcal{C}))$.

A.4 The Riesz Representation Theorem for L^2

In this section, we build on concepts developed in Chapter 5 where normed vector spaces were investigated, and establish a useful result about *linear functionals*. Since general linear functionals are not explicitly defined in the main text prior to this, we establish the definitions and the necessary geometric lemma for L^2 spaces.

Definition a.22 (Bounded Linear Functional). Let $H = L^2(\Omega, \mathcal{F}, \mu)$. A mapping $T : H \rightarrow \mathbb{R}$ is a *linear functional* if $T(af + bg) = aT(f) + bT(g)$ for all $f, g \in H$ and $a, b \in \mathbb{R}$. T is *bounded* if there exists a constant $M < \infty$ such that

$$|T(f)| \leq M \|f\|_2 \quad \forall f \in H.$$

Definition a.23 (Kernel). For a linear functional $T : H \rightarrow \mathbb{R}$, its *kernel* is the set of all elements mapped to zero:

$$\ker(T) \triangleq \{f \in H : T(f) = 0\}.$$

Lemma a.24. *A linear functional T on H is continuous if and only if it is bounded.*

CruX. ($\Leftarrow$) Let T be bounded with constant M , and let $f_n \rightarrow f$ in H . By linearity, $|T(f_n) - T(f)| = |T(f_n - f)| \leq M \|f_n - f\|_2$. Since $\|f_n - f\|_2 \rightarrow 0$, we have $T(f_n) \rightarrow T(f)$, proving continuity.

($\Rightarrow$) Assume T is continuous at the origin $0 \in H$. For $\epsilon = 1$, there exists a $\delta > 0$ such that $\|h\|_2 \leq \delta \implies |T(h)| \leq 1$. For any arbitrary non-zero $f \in H$, define the scaled element $h = \frac{\delta}{\|f\|_2} f$. Since $\|h\|_2 = \delta$, continuity guarantees $|T(h)| \leq 1$. Using the linearity of T to pull out the scalar:

$$\left| T\left(\frac{\delta}{\|f\|_2} f\right) \right| = \frac{\delta}{\|f\|_2} |T(f)| \leq 1$$

Rearranging this inequality yields:

$$|T(f)| \leq \frac{1}{\delta} \|f\|_2$$

Setting $M = 1/\delta$ provides the required finite constant, proving T is bounded. $\square$

Lemma a.25. *If T is a bounded linear functional on H , $\ker(T)$ is a complete subspace. If $T \neq 0$, there exists a non-zero $h \in H$ orthogonal to $\ker(T)$.*

CruX. Let $\{f_n\}$ be a Cauchy sequence in $\ker(T)$. Since H is complete, $f_n \rightarrow f$ for some $f \in H$. By Lemma a.24, the bounded linear functional T is continuous, meaning $T(f_n) \rightarrow T(f)$. Since $T(f_n) = 0$ for all n , $T(f) = 0$, so $f \in \ker(T)$. Thus, $\ker(T)$ contains its limits and is a complete subspace.

Since $T \neq 0$, $\ker(T)$ is a proper subspace, allowing us to choose $u \in H \setminus \ker(T)$. By Theorem 5.15 (Orthogonal Projection), there exists a unique projection $u' \in \ker(T)$ such that $h = u - u'$ is orthogonal to every element in $\ker(T)$. Since $u \notin \ker(T)$, h is non-zero. $\square$

Theorem a.26 (Riesz Representation Theorem for L^2). *Let T be a bounded linear functional on $H = L^2(\Omega, \mathcal{F}, \mu)$. There exists a unique $g \in H$ such that for all $f \in H$,*

$$T(f) = \int_{\Omega} f g \, d\mu$$

CruX. If $T = 0$, choose $g = 0$. If $T \neq 0$, Lemma a.24 provides a non-zero $h \perp \ker(T)$. Define the scaled element:

$$g = \frac{T(h)}{\|h\|_2^2} h$$

For any $f \in H$, we can decompose it as:

$$f = \left(f - \frac{T(f)}{T(h)} h \right) + \frac{T(f)}{T(h)} h$$

Applying T to the first term yields $T(f) - \frac{T(f)}{T(h)}T(h) = 0$, meaning this term belongs to $\ker(T)$. Since g is a scalar multiple of h , it is entirely orthogonal to $\ker(T)$. Taking the inner product of f with g annihilates the first term:

$$\int_{\Omega} fg \, d\mu = 0 + \int_{\Omega} \left(\frac{T(f)}{T(h)} h \right) g \, d\mu = \frac{T(f)}{T(h)} \frac{T(h)}{\|h\|_2^2} \int_{\Omega} h^2 \, d\mu = T(f) \frac{\|h\|_2^2}{\|h\|_2^2} = T(f)$$

For uniqueness, if $g_1, g_2 \in H$ both represent T , then $\int_{\Omega} f(g_1 - g_2) \, d\mu = 0$ for all $f \in H$. Setting the test function $f = g_1 - g_2$ yields $\|g_1 - g_2\|_2^2 = 0$, meaning $g_1 = g_2$ almost everywhere. $\square$